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Ahmed System for Cooperative Addition (ASCA)

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ASCA Mathematical Specification v1.0

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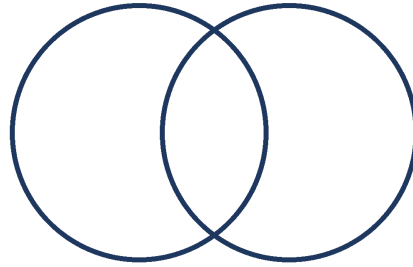


Figure 1 — ASCA Cooperative Addition Glyph

Abstract

The Ahmed System for Cooperative Addition (ASCA) is a finite mathematical operation defined on the input set $D=\{0,1,2\}$. The operation maps every ordered input pair in $D \times D$ to one of six output values. Its defining table is symmetric about the principal diagonal, and its output structure uniquely identifies the unordered pair of inputs.

This document provides the formal definition of ASCA, its truth table, its principal structural properties, theorems and corollaries derived from the table, its canonical decomposition, and the specification of its dedicated graphical glyph.

The present document concerns the finite ASCA system defined below and does not claim a generalization beyond the stated domain and definition.

1. Formal Definition

Let $D=\{0,1,2\}$ and let the output set be $R=\{-1,1,2,11,21,22\}$. ASCA is the function $f:D \times D \rightarrow R$ defined by the operation $a \oplus b = f(a,b)$. The complete definition of the operation is given by its truth table.

2. ASCA Truth Table

$\oplus$	0	1	2
0	-1	1	2
1	1	11	21
2	2	21	22

3. Finite Output Range

The complete range of ASCA is $R=\{-1,1,2,11,21,22\}$. Every element of R occurs as an output. Thus $f(D \times D)=R$. Since $R \neq D$, the term output range is used rather than closure. $|D|=3$, $|D \times D|=9$, and $|R|=6$.

4. Commutativity

The ASCA operation is commutative: $a \oplus b = b \oplus a$ for all $a,b \in D$. This follows directly from the symmetry of the truth table about the principal diagonal.

5. Zero Pass-Through

For the nonzero elements of D , $0 \oplus x = x$ for $x \in \{1,2\}$. Explicitly, $0 \oplus 1 = 1$ and $0 \oplus 2 = 2$. However, $0 \oplus 0 = -1$. Therefore, 0 is not a global identity element.

6. Negative Output Characterization

The only negative output of ASCA is -1. It occurs exclusively for the input pair (0,0). Therefore $f(a,b) < 0$ if and only if $a=0$ and $b=0$. Equivalently, $f(a,b) = -1$ if and only if $(a,b) = (0,0)$.

7. Strict Monotonicity in Each Coordinate

ASCA is strictly increasing with respect to either input when the other input is held fixed. For every fixed $b \in D$, $a < c$ implies $a \oplus b < c \oplus b$. Likewise, for every fixed $a \in D$, $b < d$ implies $a \oplus b < a \oplus d$.

8. Product-Order Monotonicity

Define the product order on $D \times D$ by $(a,b) \leq_{\text{prod}} (c,d)$ if and only if $a \leq c$ and $b \leq d$. If $(a,b) \leq_{\text{prod}} (c,d)$, then by coordinatewise monotonicity we have $a \oplus b \leq c \oplus b$ and $c \oplus b \leq c \oplus d$. By transitivity of $\leq$, it follows that $a \oplus b \leq c \oplus d$.

9. Cancellation

ASCA is cancellative in each coordinate. For fixed a , $a \oplus b = a \oplus c$ implies $b = c$. Likewise, for fixed b , $a \oplus b = c \oplus b$ implies $a = c$. This follows because every row and every column of the ASCA truth table contains distinct output values.

10. Surjectivity onto the Defined Output Set

Every element of R is produced: $-1 = 0 \oplus 0$, $1 = 0 \oplus 1$, $2 = 0 \oplus 2$, $11 = 1 \oplus 1$, $21 = 1 \oplus 2$, and $22 = 2 \oplus 2$. Therefore $f(D \times D) = R$, so f is surjective onto its defined output set R .

11. Equality Characterization — Theorem 1

For any $a, b, c, d \in D$, $f(a,b) = f(c,d)$ if and only if $\{a,b\} = \{c,d\}$.

Proof. Because ASCA is commutative, $f(a,b) = f(b,a)$. The three diagonal input pairs $(0,0), (1,1), (2,2)$ produce the distinct outputs $-1, 11, 22$. The three unordered non-diagonal pairs $\{0,1\}, \{0,2\}, \{1,2\}$ produce the distinct outputs $1, 2, 21$. Moreover, the two output sets $\{-1, 11, 22\}$ and $\{1, 2, 21\}$ are disjoint. Therefore no diagonal pair can share an output with a non-diagonal pair, and the only possible equality between two ordered input pairs is the reversal of a distinct pair. Hence $f(a,b) = f(c,d)$ exactly when $\{a,b\} = \{c,d\}$. ■

12. Fiber and Cardinality Structure — Corollary 1

For every output $y \in R$, the cardinality of its fiber is 1 when the corresponding input pair is diagonal and 2 when it is non-diagonal.

$$\begin{aligned} f^{-1}(-1) &= \{(0,0)\} \\ f^{-1}(1) &= \{(0,1), (1,0)\} \\ f^{-1}(2) &= \{(0,2), (2,0)\} \\ f^{-1}(11) &= \{(1,1)\} \\ f^{-1}(21) &= \{(1,2), (2,1)\} \\ f^{-1}(22) &= \{(2,2)\} \end{aligned}$$

Therefore $9 = 1 + 2 + 2 + 1 + 2 + 1$.

13. Recoverability

The ASCA output uniquely determines the unordered input pair. For example, 21 determines $\{1,2\}$, while 11 determines $\{1,1\}$. The ordering of distinct inputs is the only information not retained: $f(1,2) = f(2,1) = 21$.

14. Unordered-Pair Bijection

Let $P = \{\{0,0\}, \{0,1\}, \{0,2\}, \{1,1\}, \{1,2\}, \{2,2\}\}$. Then $|P|=6$. Define $f: P \rightarrow R$ by $f(\{a,b\}) = f(a,b)$. By Theorem 1, f is injective. Since $|P|=|R|=6$ and f is surjective onto R , f is a bijection.

15. Absence of a Global Identity Element

A global identity $e \in D$ would require $e \oplus x = x$ for every $x \in D$. No such element exists. For $e=0$, $0 \oplus 0 = -1 \neq 0$. For $e=1$, $1 \oplus 1 = 11 \neq 1$. For $e=2$, $2 \oplus 2 = 22 \neq 2$.

16. Absence of Idempotent Elements

An element $x \in D$ is idempotent if $x \oplus x = x$. For ASCA, $0 \oplus 0 = -1$, $1 \oplus 1 = 11$, and $2 \oplus 2 = 22$. Therefore no idempotent element exists.

17. Associativity

The expression $(a \oplus b) \oplus c$ is not generally defined within ASCA v1.0 because the first operation may produce an element outside the input domain D . For example, $1 \oplus 2 = 21$, but $21 \notin D$. Therefore associativity is not classified as either true or false for ASCA v1.0. It is not applicable under the present domain definition. This does not constitute a proof of non-associativity; it reflects the fact that the operation is not closed on its input domain.

18. Distributivity

Distributivity normally concerns the interaction of two operations. ASCA v1.0 defines only one binary operation, $\oplus$. No second operation has been defined against which distributivity could be evaluated. Therefore distributivity is not applicable to ASCA v1.0 as currently defined.

19. Canonical Decomposition

The ASCA mapping can be described structurally through three stages.

Stage 1 — Zero Case: If $(a,b) = (0,0)$, then $f(a,b) = -1$. If exactly one input is zero, then $f(0,x) = f(x,0) = x$ for $x \in \{1,2\}$.

Stage 2 — Canonicalization: For nonzero inputs, the ordered pair (a,b) is reduced to the unordered canonical pair $(\max(a,b), \min(a,b))$.

Stage 3 — Encoding: The canonical pair is represented by the corresponding ASCA output.

Thus: Input Pair $\rightarrow$ Canonical Unordered Pair $\rightarrow$ ASCA Output, with the exceptional zero-zero case represented by -1.

20. Equivalence-Class Structure

Define a relation $\sim$ on $D \times D$ by $(a,b) \sim (c,d)$ if and only if $f(a,b) = f(c,d)$. By Theorem 1, $(a,b) \sim (c,d)$ if and only if $\{a,b\} = \{c,d\}$. Therefore the six equivalence classes are exactly the fibers listed above. There are six equivalence classes, each corresponding to exactly one ASCA output.

21. Structural Summary

ASCA has the following established properties: finite domain; finite output range; commutativity; zero pass-through; unique negative output; strict monotonicity in each coordinate; product-order monotonicity; cancellation; surjectivity; Equality Characterization; Fiber/Cardinality Structure; Recoverability; unordered-pair bijection; no global identity; no idempotent elements; canonical decomposition; and six equivalence classes under output equality.

22. ASCA Cooperative Addition Glyph

The official graphical symbol associated with ASCA is designated the ASCA Cooperative Addition Glyph. It consists of two congruent hollow circles whose circumferences overlap symmetrically, producing a central Vesica Piscis region. The circles have no interior fill, shading, central axis line, or other internal graphical element. The glyph is intended to represent the cooperative combination of two inputs into one ASCA output.

The glyph's geometric symmetry is conceptually compatible with the algebraic property $a \oplus b = b \oplus a$. This correspondence is a design relationship and is not itself a mathematical proof of commutativity.

The glyph shown in Figure 1 is the graphical symbol included in ASCA v1.0.

23. Scope and Limitations

ASCA v1.0 is defined specifically on $D=\{0,1,2\}$. The present specification does not assert that the same operation, output encoding, or structural properties automatically extend to a larger numerical domain. Any generalized version of ASCA must be defined independently and subjected to its own mathematical verification.

The present document does not claim that ASCA constitutes a new branch of mathematics or a new general algebraic theory. It documents the formally defined finite system and the properties derived from that definition.

24. Version Statement

This document defines ASCA v1.0 as the finite system specified by the domain, operation table, output range, and structural properties contained herein. Subsequent modifications to the mathematical definition, domain, output encoding, or fundamental operation shall constitute a new version and shall be documented separately.

25. Author and Contact

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26. Final Statement

The Ahmed System for Cooperative Addition (ASCA) is formally specified by the finite operation $f:D \times D \rightarrow R$ with $D=\{0,1,2\}$ and $R=\{-1,1,2,11,21,22\}$. The defining table, together with the structural results established in this document, completely specifies ASCA v1.0.

The central structural result is the Equality Characterization $f(a,b)=f(c,d)$ if and only if $\{a,b\}=\{c,d\}$, which establishes that each output uniquely identifies an unordered input pair.

Accordingly, the six outputs of ASCA correspond bijectively to the six unordered pairs with repetition drawn from $\{0,1,2\}$.

This document constitutes the formal mathematical specification of ASCA v1.0 for publication on 14 August 2026.